Study of the out-of-equilibrium dynamics of one-dimensional Bose gases

Étude de la dynamique hors équilibre des gaz de Bosons unidimensionnels

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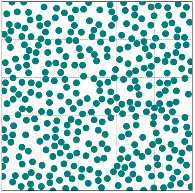
github.com/THEMEZE

Rapidities and GGE in a classical system

Chaotic/Ergodic

Integrable/non-ergidic



particule density

n

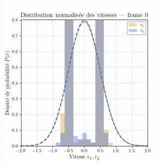
nu ← velocity

ε

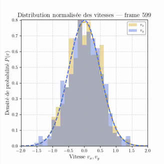
$\rho(v)$

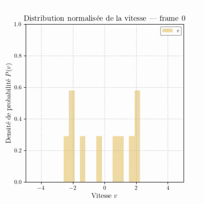
energie density

velocity density



thermalisation $\Longrightarrow$





Thesis Problematic

Scientific Question

How can we characterize the spatio-temporal evolution of a 1D quantum gas after a perturbation, in terms of the local rapidity distribution and its link to Generalized Gibbs Ensembles (GGE)?

Objectives

- Lieb and Liniger;
- GHD;
- Manip;
- Bipartition;
- Fluctuation

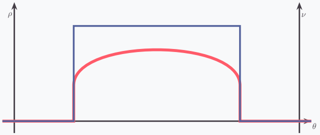
The N-body problem: the scattering phase

Kinetic energy

Contact potential

$$\hat{H} = -\frac{\hbar^2}{2m}\sum_{a=1}^{N} \hat{\partial}_{x_a}^2 + g\sum_{1\leqslant a<b\leqslant N}\delta(\hat{x}_a-\hat{x}_b)$$

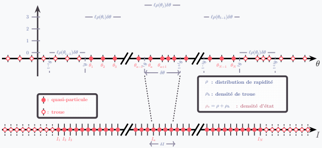
1D repulsion strength



$\{\theta_1,\cdots,\theta_N\}$

$$\psi_{\{\theta_a\}}(x_1<\cdots<x_N) \propto \sum_{\text{perm. }\sigma} e^{i\Phi\,\sigma(\{\theta_a\})}\, e^{i\frac{m}{\hbar}\sum_{b=1}^{N}\theta_{\sigma(b)}x_b}$$

$|\{\theta_a\}\rangle$



$$\frac{m}{\hbar}\theta_a L + \sum_{b=1}^{N}\Phi(\theta_a-\theta_b) = 2\pi I_a \quad a=1,\cdots,N$$

$$\left\{\begin{array}{llll} L\rho(\theta)\delta\theta & :\# & \text{rapidities} & \in[\theta,\theta+\delta\theta] \\ L\rho_h(\theta)\delta\theta & :\# & \text{holes} & \in[\theta,\theta+\delta\theta] \\ L\rho_s(\theta)\delta\theta & :\# & \text{states} & \in[\theta,\theta+\delta\theta] \end{array}\right.$$

The N-body problem: the scattering phase

Kinetic energy

Contact potential

$$\hat{H} = -\frac{\hbar^2}{2m}\sum_{a=1}^{N}\hat{\partial}^2_{x_a} + g\sum_{1\leqslant a<b\leqslant N}\delta\left(\hat{x}_a-\hat{x}_b\right)$$

1D repulsion strength

$\{\theta_1,\cdots,\theta_N\}$

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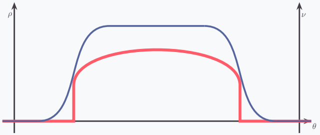

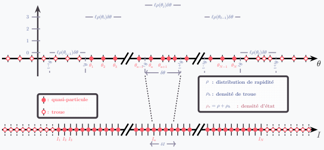

$$\begin{cases} L\rho(\theta)\delta\theta & : \#\ \text{rapidities} \in [\theta,\theta+\delta\theta] \\ L\rho_h(\theta)\delta\theta & : \#\ \text{holes} \in [\theta,\theta+\delta\theta] \\ L\rho_s(\theta)\delta\theta & : \#\ \text{states} \in [\theta,\theta+\delta\theta] \end{cases}$$

Comparaison des commandes

- **Uncover** : espace réservé

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- **Uncover** : espace réservé
- **Visible** : sans espace réservé

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- **Onslide** : garde l'espace

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- **Onslide** : garde l'espace
- A : alterne contenu

A frame with title and subtitle

Subtitle here

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua

Itemized list:

- Lorem ipsum
- Dolor sit amet
  - Consectetur
  - Adipiscing elit
- Sed do eiusmod
  - Tempor incididunt
    - Ut labore et dolore
    - Magna aliqua

A frame with title only

Theorem

$$e^{i\pi} + 1 = 0$$

Démonstration

$$e^{iz} = \cos z + i \sin z$$

therefore

$$\begin{aligned} e^{i\pi} + 1 &= \cos \pi + i \sin \pi + 1 \\ &= -1 + i \times 0 + 1 \\ &= 0 \end{aligned}$$

A frame with background image

You can still add title and subtitle.

You can also use a background in the title slide by setting:

```
\frame[plain,bg=demo-background.jpg]{\titlepage}
```

A plain frame has no headline

`\Alegreya`	Lorem ipsum dolor sit amet
`\AlegreyaMedium`	Lorem ipsum dolor sit amet
`\AlegreyaExtraBold`	**Lorem ipsum dolor sit amet**
`\AlegreyaBlack`	**Lorem ipsum dolor sit amet**
`\AlegreyaSansThin`	Lorem ipsum dolor sit amet
`\AlegreyaSansLight`	Lorem ipsum dolor sit amet
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`\AlegreyaSansMedium`	Lorem ipsum dolor sit amet
`\AlegreyaSansExtraBold`	**Lorem ipsum dolor sit amet**
`\AlegreyaSansBlack`	**Lorem ipsum dolor sit amet**

Alert! A *plain* frame does not show the progress bar but still appears in it unless the frame comes after \End

A ***STANDOUT*** frame can be used to focus attention

In combination with *plain*,
it makes a nice thank-you slide!

https://github.com/piazzai/arguelles
https://ctan.org/pkg/beamertheme-arguelles